MSG No. 191.236 $P6'/mm'm$ [Type III, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6'/mm'm`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `6'/mm'm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `2c`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,7,8,9,16,17,18,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[4,5,6,10,11,12,13,14,15,19,20,21]

Table 4: Wyckoff site: `2d`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,7,8,9,16,17,18,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,5,6,10,11,12,13,14,15,19,20,21]

Table 5: Wyckoff site: `2e`, site symmetry: `6'm'm`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$[0, 0, -z]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 6: Wyckoff site: `3f`, site symmetry: `mm'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,8,11,14,17,19,22]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,5,9,12,15,18,20,23]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,6,7,10,13,16,21,24]

Table 7: Wyckoff site: `3g`, site symmetry: `mm'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,8,11,14,17,19,22]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,9,12,15,18,20,23]

3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,6,7,10,13,16,21,24]`

Table 8: Wyckoff site: `4h`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,22,23,24]`
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[4,5,6,19,20,21]`
3	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[10,11,12,13,14,15]`

Table 9: Wyckoff site: `6i`, site symmetry: `2'm'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,11,14,22]`
2	$\left[0, \frac{1}{2}, z\right]$	`[2,12,15,23]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,10,13,24]`
4	$\left[\frac{1}{2}, 0, -z\right]$	`[4,8,17,19]`
5	$\left[0, \frac{1}{2}, -z\right]$	`[5,9,18,20]`
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6,7,16,21]`

Table 10: Wyckoff site: `6j`, site symmetry: `m2m`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,8,11]`
2	$[0, x, 0]$	`[2,5,9,12]`
3	$[-x, -x, 0]$	`[3,6,7,10]`
4	$[x, x, 0]$	`[13,16,21,24]`
5	$[-x, 0, 0]$	`[14,17,19,22]`
6	$[0, -x, 0]$	`[15,18,20,23]`

Table 11: Wyckoff site: `6k`, site symmetry: `m2m`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4,8,11]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,5,9,12]`
3	$\left[-x, -x, \frac{1}{2}\right]$	`[3,6,7,10]`
4	$\left[x, x, \frac{1}{2}\right]$	`[13,16,21,24]`
5	$\left[-x, 0, \frac{1}{2}\right]$	`[14,17,19,22]`
6	$\left[0, -x, \frac{1}{2}\right]$	`[15,18,20,23]`

Table 12: Wyckoff site: `6l`, site symmetry: `mm'2'`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,8,17,22]`
2	$[-2x,\ -x,\ 0]$	`[2,9,18,23]`
3	$[x,\ -x,\ 0]$	`[3,7,16,24]`
4	$[-x,\ -2x,\ 0]$	`[4,11,14,19]`
5	$[2x,\ x,\ 0]$	`[5,12,15,20]`
6	$[-x,\ x,\ 0]$	`[6,10,13,21]`

Table 13: Wyckoff site: `6m`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,8,17,22]`
2	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[2,9,18,23]`
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,7,16,24]`
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,11,14,19]`
5	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[5,12,15,20]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,10,13,21]`

Table 14: Wyckoff site: `12n`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,11]`
2	$[0,\ x,\ z]$	`[2,12]`
3	$[-x,\ -x,\ z]$	`[3,10]`
4	$[x,\ 0,\ -z]$	`[4,8]`
5	$[0,\ x,\ -z]$	`[5,9]`
6	$[-x,\ -x,\ -z]$	`[6,7]`
7	$[x,\ x,\ z]$	`[13,24]`
8	$[-x,\ 0,\ z]$	`[14,22]`
9	$[0,\ -x,\ z]$	`[15,23]`
10	$[x,\ x,\ -z]$	`[16,21]`
11	$[-x,\ 0,\ -z]$	`[17,19]`
12	$[0,\ -x,\ -z]$	`[18,20]`

Table 15: Wyckoff site: `12o`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,22]`
2	$[-2x,\ -x,\ z]$	`[2,23]`
3	$[x,\ -x,\ z]$	`[3,24]`
4	$[-x,\ -2x,\ -z]$	`[4,19]`
5	$[2x,\ x,\ -z]$	`[5,20]`
6	$[-x,\ x,\ -z]$	`[6,21]`
7	$[x,\ -x,\ -z]$	`[7,16]`

continued ...

Table 15

No.	position	mapping
8	$[x,\ 2x,\ -z]$	`[8,17]`
9	$[-2x,\ -x,\ -z]$	`[9,18]`
10	$[-x,\ x,\ z]$	`[10,13]`
11	$[-x,\ -2x,\ z]$	`[11,14]`
12	$[2x,\ x,\ z]$	`[12,15]`

Table 16: Wyckoff site: `12p`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[-y,\ x-y,\ 0]$	`[2,9]`
3	$[-x+y,\ -x,\ 0]$	`[3,7]`
4	$[x-y,\ -y,\ 0]$	`[4,11]`
5	$[y,\ x,\ 0]$	`[5,12]`
6	$[-x,\ -x+y,\ 0]$	`[6,10]`
7	$[x-y,\ x,\ 0]$	`[13,21]`
8	$[-x,\ -y,\ 0]$	`[14,19]`
9	$[y,\ -x+y,\ 0]$	`[15,20]`
10	$[x,\ x-y,\ 0]$	`[16,24]`
11	$[-x+y,\ y,\ 0]$	`[17,22]`
12	$[-y,\ -x,\ 0]$	`[18,23]`

Table 17: Wyckoff site: `12q`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,8]`
2	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[2,9]`
3	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[3,7]`
4	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[4,11]`
5	$\left[y,\ x,\ \frac{1}{2}\right]$	`[5,12]`
6	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[6,10]`
7	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[13,21]`
8	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[14,19]`
9	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	`[15,20]`
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[16,24]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[17,22]`
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[18,23]`

Table 18: Wyckoff site: `24r`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`

continued ...

Table 18

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x-y,\ -y,\ -z]$	[4]
5	$[y,\ x,\ -z]$	[5]
6	$[-x,\ -x+y,\ -z]$	[6]
7	$[-x+y,\ -x,\ -z]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$[-y,\ x-y,\ -z]$	[9]
10	$[-x,\ -x+y,\ z]$	[10]
11	$[x-y,\ -y,\ z]$	[11]
12	$[y,\ x,\ z]$	[12]
13	$[x-y,\ x,\ z]$	[13]
14	$[-x,\ -y,\ z]$	[14]
15	$[y,\ -x+y,\ z]$	[15]
16	$[x,\ x-y,\ -z]$	[16]
17	$[-x+y,\ y,\ -z]$	[17]
18	$[-y,\ -x,\ -z]$	[18]
19	$[-x,\ -y,\ -z]$	[19]
20	$[y,\ -x+y,\ -z]$	[20]
21	$[x-y,\ x,\ -z]$	[21]
22	$[-x+y,\ y,\ z]$	[22]
23	$[-y,\ -x,\ z]$	[23]
24	$[x,\ x-y,\ z]$	[24]